WORKED SOLUTION

STEP I 2000, Question 3

Pure • Integrals of Step Functions

STEP I 2000, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q3

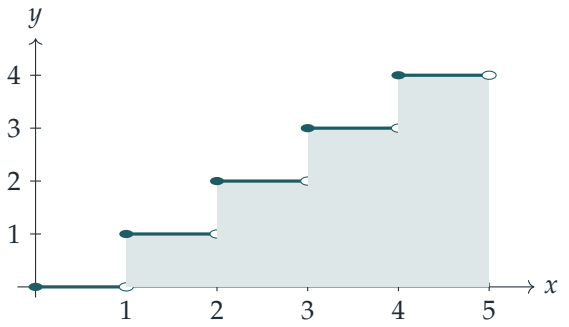
The key idea

The integer part of anything is a step function: it stays constant on intervals and then jumps by one. So the integral is just a sum of rectangle areas, and the only real work is finding where each step starts and stops. For $[\mathrm{e}^x]$ the jumps happen where e^x passes an integer, that is, at $x = \ln k$, and the resulting sum of logarithms telescopes neatly.

Solution

The graph of $y = [x]$

On each interval $k \leq x < k+1$ (with k an integer) we have $[x] = k$. So the graph is a staircase of horizontal segments, each of length 1, with a filled dot at the left end (the value is attained) and an open dot at the right end (where the function jumps up by 1).



The integral is the shaded area, made of rectangles of width 1 and heights $0, 1, 2, 3, 4$ (the value at the single points $x = 1, 2, 3, 4$ makes no difference to the area). Hence

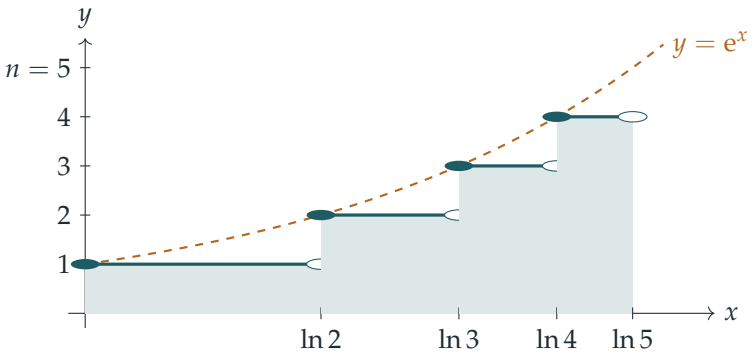
$$\int_0^5 [x]\,\mathrm{d}x = 0 + 1 + 2 + 3 + 4 = 10.$$

The graph of $y = [\mathrm{e}^x]$

For $0 \le x < \ln n$ we have $1 \le \mathrm{e}^x < n$. Since e^x is increasing,

$$[\mathrm{e}^x] = k \iff k \le \mathrm{e}^x < k+1 \iff \ln k \le x < \ln(k+1),$$

for $k = 1, 2, \ldots, n-1$. So the graph is again a staircase with steps of height 1, but the steps now get shorter as x increases: the step at height k has width $\ln(k+1) - \ln k = \ln(1 + \frac{1}{k})$. The sketch shows $n = 5$, with the curve $y = \mathrm{e}^x$ dashed; the staircase touches the curve at the left end of each step.



The integral

Adding the rectangle areas, height k times width $\ln(k+1) - \ln k$,

$$\int_0^{\ln n} [\mathrm{e}^x]\,\mathrm{d}x = \sum_{k=1}^{n-1} k(\ln(k+1) - \ln k) = \sum_{k=1}^{n-1} k\ln(k+1) - \sum_{k=1}^{n-1} k\ln k.$$

? **Ask yourself** How do we make the two sums cancel?

Each $\ln j$ appears in both sums, with different coefficients. Substituting $j = k+1$ in the first sum lines them up:

$$\begin{aligned}\sum_{k=1}^{n-1} k\ln(k+1) - \sum_{k=1}^{n-1} k\ln k &= \sum_{j=2}^{n}(j-1)\ln j - \sum_{j=1}^{n-1} j\ln j \\ &= (n-1)\ln n + \sum_{j=2}^{n-1}((j-1)-j)\ln j - 1\cdot\ln 1 \\ &= (n-1)\ln n - \sum_{j=2}^{n-1}\ln j.\end{aligned}$$

Here we took the $j = n$ term out of the first sum, and the $j = 1$ term out of the second (it is zero, since $\ln 1 = 0$). Now $\sum_{j=2}^{n-1}\ln j = \ln((n-1)!)$, and $(n-1)\ln n = n\ln n - \ln n$, so

$$\int_0^{\ln n} [\mathrm{e}^x]\,\mathrm{d}x = n\ln n - (\ln n + \ln((n-1)!)) = n\ln n - \ln(n!),$$

as required. (For $n = 1$ the integral is over an interval of length zero, and both sides are 0.)

A picture-proof (optional)

The rectangle of width $\ln n$ and height n has area $n \ln n$. The part of it lying *above* the staircase is made of horizontal strips of height 1: between heights $m-1$ and m, for $m = 1, \ldots, n$, the strip runs from $x = 0$ to $x = \ln m$, because $[\mathrm{e}^x] < m$ exactly when $x < \ln m$. Its area is $\ln m$, so the region above the steps has area $\ln 1 + \ln 2 + \cdots + \ln n = \ln(n!)$. Subtracting from $n \ln n$ gives the result again. This is the discrete version of integrating with respect to y instead of x.

Checks

For $n = 2$ the staircase is a single rectangle of height 1 and width $\ln 2$, and the formula gives $2\ln 2 - \ln 2 = \ln 2$. For $n = 4$ the formula gives $4\ln 4 - \ln 24 \approx 2.3671$; a midpoint-rule evaluation of $\int_0^{\ln 4} [\mathrm{e}^x]\,\mathrm{d}x$ agrees to five decimal places, and so it does for every n up to 8. The answer must also be less than $\int_0^{\ln n} \mathrm{e}^x\,\mathrm{d}x = n - 1$, since the steps sit under the curve: for $n = 4$, $2.37 < 3$.

Common slips

- Taking the step at height k to have width 1 (copying the first part), rather than $\ln(k+1) - \ln k$.
- Starting the staircase at height 0: on $0 \le x < \ln n$ we have $\mathrm{e}^x \ge 1$, so the lowest step is at height 1.
- Mis-shifting the index and losing the $(n-1)\ln n$ term, or not noticing it must combine with $\ln((n-1)!)$ to give $\ln(n!)$.

Beyond STEP

Since the steps lie under $y = \mathrm{e}^x$, the integral is less than $n - 1$, which rearranges to $\ln(n!) > n\ln n - n + 1$. Bounding the steps from above in the same way leads towards Stirling's formula, $\ln(n!) \approx n\ln n - n$, a key estimate in analysis, probability and combinatorics. The swap from a sum of rectangles to a sum of strips is also an example of "summation by parts", the discrete cousin of integration by parts. Abbott's *Understanding Analysis* is a good next step for putting integrals of step functions on a rigorous footing.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.